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$$\vec{A}-\vec{B}=(A_x-B_x)\hat{i}+(A_y-B_y)\hat{j}+(A_z-B_z)\hat{k}$$

The above rules form an analytical way of adding and subtracting two vectors.

EXAMPLE 2.4

Two vectors $\vec{A}$ and $\vec{B}$ are given in the component form as $\vec{A}=5\hat{i}+7\hat{j}-4\hat{k}$ and $\vec{B}=6\hat{i}+3\hat{j}+2\hat{k}$. Find $\vec{A}+\vec{B}$, $\vec{B}+\vec{A}$, $\vec{A}-\vec{B}$, $\vec{B}-\vec{A}$

Solution

$$\vec{A}+\vec{B}=(5\hat{i}+7\hat{j}-4\hat{k})+(6\hat{i}+3\hat{j}+2\hat{k})$$
$$=11\hat{i}+10\hat{j}-2\hat{k}$$
$$\vec{B}+\vec{A}=(6\hat{i}+3\hat{j}+2\hat{k})+(5\hat{i}+7\hat{j}-4\hat{k})$$
$$=(6+5)\hat{i}+(3+7)\hat{j}+(2-4)\hat{k}$$
$$=11\hat{i}+10\hat{j}-2\hat{k}$$
$$\vec{A}-\vec{B}=(5\hat{i}+7\hat{j}-4\hat{k})-(6\hat{i}+3\hat{j}+2\hat{k})$$
$$=-\hat{i}+4\hat{j}-6\hat{k}$$
$$\vec{B}-\vec{A}=\hat{i}-4\hat{j}+6\hat{k}$$

Note that the vectors $\vec{A}+\vec{B}$ and $\vec{B}+\vec{A}$ are same and the vectors $\vec{A}-\vec{B}$ and $\vec{B}-\vec{A}$ are opposite to each other.

Note

The addition of two vectors using components depends on the choice of the coordinate system. But the geometric way of adding and subtracting two vectors is independent of the coordinate system used.

2.5 MULTIPLICATION OF VECTOR BY A SCALAR

A vector $\vec{A}$ multiplied by a scalar λ results in another vector, $\lambda\vec{A}$. If λ is a positive number then $\lambda\vec{A}$ is also in the direction of $\vec{A}$. If λ is a negative number, $\lambda\vec{A}$ is in the opposite direction to the vector $\vec{A}$.

EXAMPLE 2.5

Given the vector $\vec{A}=2\hat{i}+3\hat{j}$, what is $3\vec{A}$?

Solution

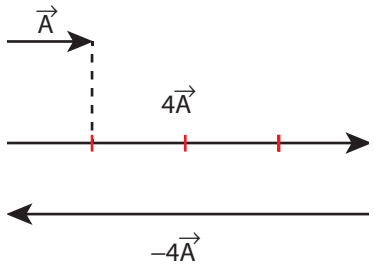
$$3\vec{A}=3\left(2\hat{i}+3\hat{j}\right)=6\hat{i}+9\hat{j}$$

The vector $3\vec{A}$ is in the same direction as vector $\vec{A}$.

EXAMPLE 2.6

A vector $\vec{A}$ is given as in the following Figure. Find $4\vec{A}$ and $-4\vec{A}$

Solution



In physics, certain vector quantities can be defined as a scalar times another vector quantity.

For example

1) Force $\vec{F}=m\vec{a}$. Here mass 'm' is a scalar, and $\vec{a}$ is the acceleration. Since 'm' is always a positive scalar, the direction